

Emotional Models: Supervised vs. Zero-Shot Classification

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1 Introduction

This project explores two distinct approaches to emotion classification in text using Transformer-based models: supervised learning with feature extraction and zero-shot classification via Natural Language Inference (NLI).

2 Dataset Description

The dataset utilized in this project is the `dair-ai/emotion` dataset. It consists of text samples, primarily collected from Twitter, labeled with one of six emotions: *joy*, *sadness*, *anger*, *fear*, *love* and *surprise*. The dataset is split into training, validation, and test sets, with the training set containing 16,000 samples, the validation set 2,000 samples, and the test set 2,000 samples. The samples are distributed according to the following table:

Table 1. Class distribution of the `dair-ai/emotion` dataset.

Emotion	Train	Validation	Test	Total (%)
Joy	5,362	704	695	33.8%
Sadness	4,666	550	581	29.0%
Anger	2,159	275	275	13.5%
Fear	1,937	212	224	11.9%
Love	1,304	178	159	8.2%
Surprise	572	81	66	3.6%
Total	16,000	2,000	2,000	100.0%

2.1 Foreseeable issues

Class imbalance By looking at table 1, we can see that the dataset is imbalanced, with the *joy* and *sadness* classes being significantly more represented, together accounting for over 60% of the data, and *surprise* being the least represented by a considerable margin, 3.6%. This imbalance could lead to biased model performance, favoring the majority classes, and already gives away that “Accuracy” might not be the best metric to evaluate the models.

Overfitting to keywords Another issue present is the overfitting of words or terms to a certain emotion. For example, the word “happy” is strongly associated with the *joy* class, and its presence in a text sample could lead the model to predict *joy* regardless of the overall context. One example of this is [por exemplo...].

3 Methodology

To address the problem of emotion classification, we implemented two distinct pipelines: a supervised learning approach using frozen Transformer embeddings and a zero-shot inference approach using a Natural Language Inference (NLI) model. All experiments were conducted using the `dair-ai/emotion` dataset.

3.1 Supervised Learning: Feature Extraction

In this approach, we leveraged a pre-trained Transformer to generate fixed semantic representations of the input text, which were then fed into a custom Artificial Neural Network (ANN) classifier.

Embedding Generation. We utilized the `sentence-transformers/all-MiniLM-L6-v2` model as our backbone. This model was chosen for its balance between performance and computational efficiency. We processed the raw text from the train, validation, and test splits to generate dense vector embeddings. The backbone weights remained frozen during this process, serving purely as a static feature extractor.

Classifier Architecture. On top of the generated embeddings, we trained a non-linear classifier (Multi-Layer Perceptron). The architecture, defined in our `ANNClassifier` module, consists of:

- **Input Layer:** Matches the dimension of the embeddings (384 dimensions for MiniLM).
- **Hidden Layer:** A fully connected layer with 256 neurons, followed by a Rectified Linear Unit (ReLU) activation function.
- **Regularization:** A Dropout layer with a probability of $p = 0.1$ to prevent overfitting.
- **Output Layer:** A final fully connected layer mapping to the 6 emotion classes.

Training Configuration. The model was trained using the Adam optimizer with a learning rate of $1e - 3$ and Cross-Entropy Loss. To ensure robust evaluation, we employed the following training protocol:

- **Batch Size:** 64 samples.

- **Early Stopping:** We monitored the Macro-F1 score on the validation set, stopping training if no improvement was observed after 3 consecutive epochs (patience=3).
- **Reproducibility:** The training and evaluation were repeated across 5 distinct random seeds (starting from seed 2025) to account for initialization variance. We report the mean and standard deviation for all supervised metrics.

3.2 Zero-Shot Classification

The second approach reformulates the classification task as a Natural Language Inference (NLI) problem, requiring no supervised training on the emotion dataset.

Model and Prompting. We employed the MoritzLaurer/bge-m3-zeroshot-v2.0 model. For each input text, we constructed hypotheses using the template:

"This text expresses {}."

where the placeholder is filled with each of the candidate labels (anger, fear, joy, love, sadness, surprise). The model computes the entailment probability for each hypothesis, and the label with the highest probability is selected as the prediction. Unlike the supervised pipeline, this method was applied directly to the test set without any parameter updates.

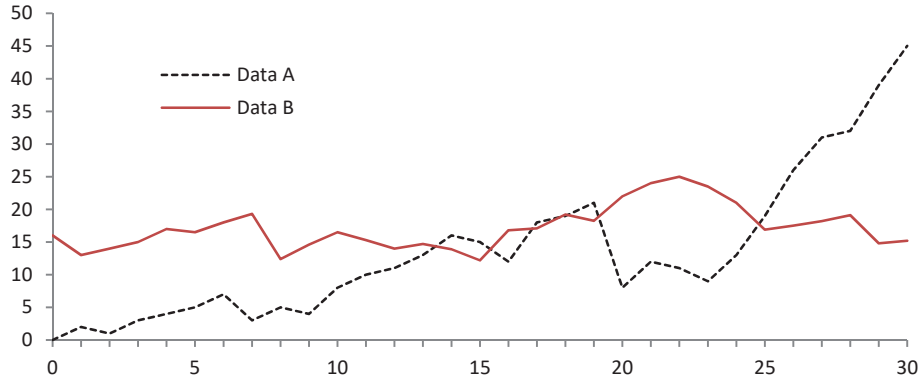


Fig. 1. A figure caption is always placed below the illustration. Please note that short captions are centered, while long ones are justified by the macro package automatically.

Theorem 1. *This is a sample theorem. The run-in heading is set in bold, while the following text appears in italics. Definitions, lemmas, propositions, and corollaries are styled the same way.*

Proof. Proofs, examples, and remarks have the initial word in italics, while the following text appears in normal font.

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [1], an LNCS chapter [2], a book [3], proceedings without editors [4], and a homepage [5]. Multiple citations are grouped [1–3], [1, 3–5].

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Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional) acknowledgments¹, for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

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